

Lecture 4: The global theory of curves

September 15

So far, curvature and torsion have described a curve near one point. Today we ask what they say about a curve as a whole.

Definition. A regular curve

$$c : [a, b] \longrightarrow \mathbb{R}^n$$

is **closed** if it extends to a smooth regular periodic curve

$$\tilde{c} : \mathbb{R} \longrightarrow \mathbb{R}^n, \quad \tilde{c}(t + b - a) = \tilde{c}(t).$$

In particular,

$$c(a) = c(b), \quad \frac{d^j c}{dt^j}(a) = \frac{d^j c}{dt^j}(b)$$

for every derivative under consideration.

The curve is **simply closed** if it has no self-intersections: the restriction of c to $[a, b)$ is injective.

Equivalently, a closed regular curve is an immersion $\mathbb{S}^1 \rightarrow \mathbb{R}^n$, and a simply closed curve is an embedding.

Definition. Let $\kappa(t) \geq 0$ denote the curvature of a regular curve $c(t)$. Its **total curvature** is

$$\int_a^b \kappa(t) \|\dot{c}(t)\| dt = \int_0^L \kappa(s) ds.$$

Here L is the length of c . The expressions $\kappa(t)$ and $\kappa(s)$ mean the same curvature function written in different parameters.

For an oriented unit-speed plane curve, choose

$$v(s) = JT(s), \quad J(x, y) = (-y, x),$$

and define the **signed curvature** $k(s)$ by

$$T'(s) = k(s)v(s).$$

Then

$$\kappa(s) = |k(s)|.$$

Thus κ always denotes the nonnegative curvature introduced in Lecture 2, while k is used only for signed curvature in the oriented plane. The vector v is the oriented plane normal, not necessarily the principal normal N . Wherever $k \neq 0$,

$$N = \text{sgn}(k)v.$$

Thus the signed total curvature is

$$\int_0^L k(s) ds,$$

while the total curvature is

$$\int_0^L |k(s)| ds.$$

Let c be a regular plane curve and write its unit tangent locally as

$$T(t) = (\cos \varphi(t), \sin \varphi(t)).$$

Then

$$\frac{dT}{dt} = \varphi'(t)(-\sin \varphi(t), \cos \varphi(t)) = \varphi'(t)v(t).$$

On the other hand,

$$\frac{dT}{dt} = \frac{dT}{ds} \frac{ds}{dt} = k(t)v(t)\|\dot{c}(t)\|.$$

Therefore

$$\boxed{k(t) = \frac{\varphi'(t)}{\|\dot{c}(t)\|}}$$

and, whenever a single angle function is defined,

$$\int_a^b k(t)\|\dot{c}(t)\| dt = \varphi(b) - \varphi(a).$$

Here $k(t)$ means the signed curvature at $c(t)$. It measures the rate at which the tangent angle turns per unit arclength.

There is one subtlety: an angle is only defined modulo 2π .

Angle lifting theorem. Let

$$\gamma : [a, b] \longrightarrow \mathbb{R}^2 \setminus \{0\}$$

be continuous. After choosing an initial angle, there is a unique continuous function $\varphi : [a, b] \rightarrow \mathbb{R}$ such that

$$\frac{\gamma(t)}{\|\gamma(t)\|} = (\cos \varphi(t), \sin \varphi(t)).$$

Any two such lifts differ by a constant multiple of 2π . Consequently,

$$\varphi(b) - \varphi(a)$$

does not depend on the chosen lift.

Idea of the proof. Subdivide $[a, b]$ so that the image of each small subinterval lies in an open half-plane through the origin. On each such half-plane the angle has a single-valued continuous branch. Starting from $\varphi(a)$, extend these branches one at a time.

Suppose now that $\gamma(a) = \gamma(b)$. Then

$$\varphi(b) - \varphi(a) \in 2\pi\mathbb{Z}.$$

Definition. The *winding number* of the closed curve γ about the origin is

$$W(\gamma, 0) = \frac{\varphi(b) - \varphi(a)}{2\pi} \in \mathbb{Z}.$$

It counts the net number of turns around the origin, with counter-clockwise turns positive and clockwise turns negative.

For example,

$$\gamma_m(t) = (\cos mt, \sin mt), \quad 0 \leq t \leq 2\pi,$$

has

$$W(\gamma_m, 0) = m.$$

The winding number is unchanged under a homotopy through closed curves which avoid the origin. It can change only if the curve passes through the origin.

The symbol

$$\int_c f ds$$

means that we integrate the quantity f *along the curve* c .

More precisely, if

$$c : [a, b] \longrightarrow \mathbb{R}^n$$

is regular, then

$$\int_c f ds := \int_a^b f(c(t)) \|\dot{c}(t)\| dt.$$

If $f(t)$ already denotes a function along the curve, we simply write

$$\int_c f ds = \int_a^b f(t) \|\dot{c}(t)\| dt.$$

The subscript c says where the integration takes place, while

$$ds = \|\dot{c}(t)\| dt$$

is the element of arclength. This factor makes the integral independent of the speed of the chosen parametrization.

For the positively oriented circle

$$c(t) = (R \cos t, R \sin t), \quad 0 \leq t \leq 2\pi,$$

we have $k = 1/R$ and $\|\dot{c}\| = R$. Hence

$$\int_c k ds = \int_0^{2\pi} \frac{1}{R} R dt = 2\pi.$$

Let $c : [a, b] \rightarrow \mathbb{R}^2$ be a regular closed curve. Its unit tangent defines a closed curve on the unit circle:

$$T : [a, b] \longrightarrow \mathbb{S}^1, \quad T(t) = \frac{\dot{c}(t)}{\|\dot{c}(t)\|}.$$

Definition. The *rotation index* of c is the winding number of its tangent:

$$U_c = W(T, 0).$$

If $T = (\cos \varphi, \sin \varphi)$, then

$$\boxed{\int_c k \, ds = \varphi(b) - \varphi(a) = 2\pi U_c.}$$

Thus the signed total curvature of every regular closed plane curve is an integer multiple of 2π .

The position vector $c(t)$ may trace a complicated path. The integer U_c records only how its tangent direction turns.

Examples show why simple closedness matters.

For the positively oriented circle,

$$U_c = 1, \quad \int_c k \, ds = 2\pi.$$

Traversing the same circle m times gives $U_c = m$, but the resulting parametrized curve is not simply closed when $|m| > 1$.

A regular figure-eight can have

$$U_c = 0.$$

One lobe turns counterclockwise and the other clockwise, so their signed contributions cancel. Its total absolute curvature is still positive:

$$\int_c |k| \, ds > \left| \int_c k \, ds \right| = 0.$$

The rotation index is unchanged under a **regular homotopy**, a deformation through regular closed curves. A cusp would allow the index to change, but at a cusp the tangent vanishes and regularity fails.